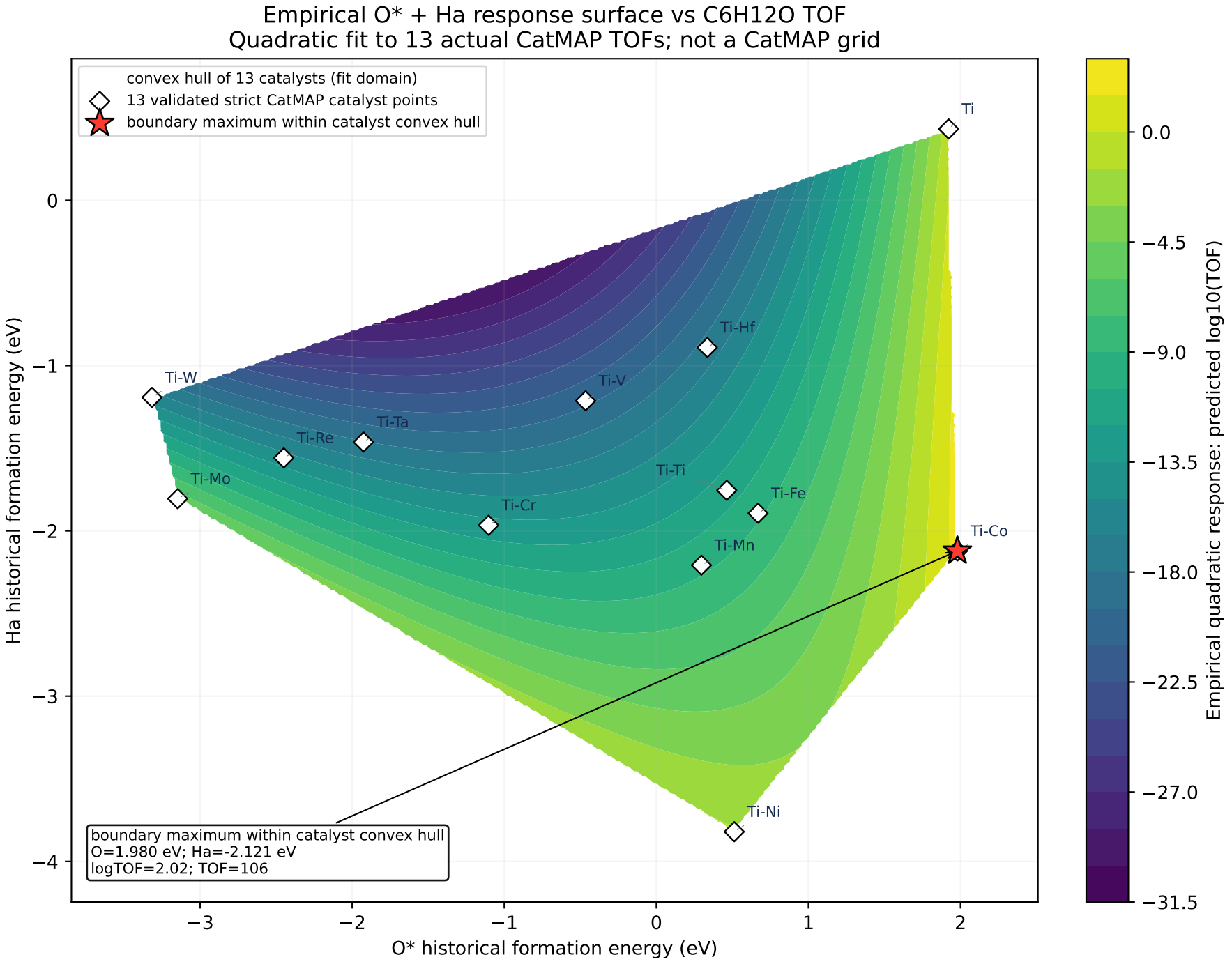

Empirical O* + Ha response surface vs C6H12O TOF
Quadratic fit to 13 actual CatMAP TOFs; not a CatMAP grid
convex hull of 13 catalysts (fit domain)
13 validated strict CatMAP catalyst points
boundary maximum within catalyst convex hull
Ti
Ti-Hf
Ti-W
Ti-V
Ti-Ta
Ti-Re
Ti-Mo
Ti-Ti
Ti-Fe
Ti-Cr
Ti-Mn
Ti-Co
Ti-Ni
boundary maximum within catalyst convex hull
O=1.980 eV; Ha=-2.121 eV
logTOF=2.02; TOF=106
Ha historical formation energy (eV)
O* historical formation energy (eV)
Empirical quadratic response: predicted log10(TOF)
0.0
−4.5
−9.0
−13.5
−18.0
−22.5
−27.0
−31.5